UAV-Enabled Passive 6DMA for Secure Communications: A Space-Nulling Approach

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***Abstract*—Intelligent reflecting surface (IRS) can be mounted on an unmanned aerial vehicle (UAV) to enhance terrestrial communication performance by leveraging its passive beamforming and the UAV's six-dimensional (6D) movement, i.e., three-dimensional (3D) rotation and 3D positioning, known as passive 6D movable antennas (6DMAs). In this paper, we investigate the use of UAV-mounted passive 6DMAs for secure communications. In particular, beyond conventional signal nulling, we propose a new space nulling scheme by leveraging the 6D movement and 180° half-space reflection of the IRS, such that eavesdroppers are excluded from the reflection space. Our goal is to maximize the achievable rate at the legitimate user subject to the space-nulling constraint at an eavesdropper by jointly optimizing the IRS's position and 3D rotation. As this problem is difficult to be optimally solved, we first derive the IRS's optimal 3D rotation in closed-form for any given UAV/IRS position by recasting it as an equivalent lower-dimensional problem. Then, we proceed to optimize the IRS's position via an exhaustive search. Both analytical and numerical results demonstrate that the proposed space-nulling scheme can achieve near-optimal performance in terms of secrecy rate maximization under certain conditions.**

I. Introduction

As one of the transformative technologies envisioned for the sixth-generation (6G) era, intelligent reflecting surface (IRS) has garnered significant attention in recent years due to its flexibility and cost-effectiveness in enhancing wireless communication performance [1]. An IRS typically comprises a large number of reflecting elements capable of dynamically altering the direction and strength of incident signals. This capability enables its application across a wide range of wireless network scenarios for different purposes, including spatial multiplexing, target sensing, interference suppression, among others [1]–[3].

However, in practice, an IRS can only reflect signals from/to its pointing half space. To enable 360° panoramic full-angle reflection, several prior works have proposed mounting IRSs on unmanned aerial vehicles (UAVs), i.e., aerial IRS (AIRS), by leveraging the higher altitude of UAVs compared to terrestrial base stations (BSs) and users [4]. Readers are referred to [5] and [6] for a comprehensive survey on the integration of IRS and UAV communications. However, most existing studies have focused only on the position optimization of the UAV/IRS, while ignoring the UAV's capability for three-dimensional (3D) posture control. This posture control offers additional degrees of freedom via 3D rotation optimization. A handful of recent works have begun to explore

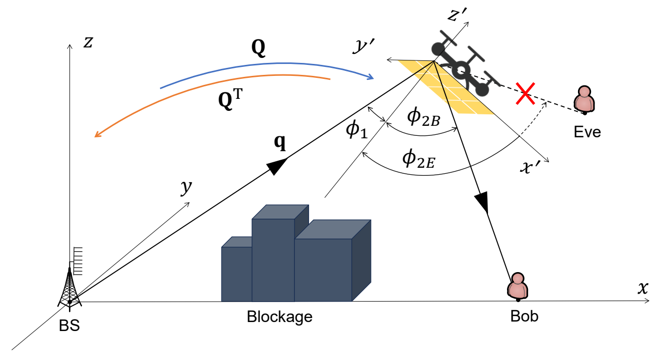

Fig. 1. Passive-6DMA-enabled space nulling for secure communications.

the joint optimization of both the position and rotation of AIRSs (also termed passive six-dimensional movable antennas (6DMA)) [8]–[12]. For example, the authors in [8] and [9] investigated joint position and one-dimensional (1D) rotation optimization for a terrestrial IRS and an AIRS, respectively. In [10] and [11], the authors focused on more general 3D rotation optimization for an AIRS in single-user and multicast systems, respectively. However, to the best of our knowledge, no existing work has investigated the use of AIRS or passive 6DMAs for secure communications. In particular, the half-space reflection of the IRS provides an efficient means to suppress eavesdropper by adjusting its orientation, such that the eavesdropper falls outside the reflection space, as depicted in Fig. 1. Notably, compared to traditional signal nulling techniques, this space-nulling strategy is more practical to implement and inherently robust against phase perturbations caused by wind disturbances in aerial environments.

To characterize the performance limit of the space-nulling strategy, we investigate an AIRS-aided secure communication system consisting of a BS, a legitimate user (Bob), and an eavesdropper (Eve), as shown in Fig. 1. Our goal is to maximize the secrecy rate by jointly optimizing the AIRS's location and 3D rotation under an angle-dependent reflection model, subject to the space-nulling constraint at Eve. However, due to the intricate coupling between position and rotation optimization, as well as the space-nulling constraint, the resulting problem is challenging to be solved optimally. To address this difficulty, we first reformulate the problem into a lower-dimensional version and derive the optimal 3D rotation in closed-form for any given AIRS location. Then, the optimal AIRS location is determined by performing an exhaustive search. Both analytical and numerical results demonstrate that the proposed space-nulling scheme can achieve near-optimal secrecy performance under specific conditions and significantly outperform other baseline schemes.

Notations: $\mathbf{x}^T$, $\|\mathbf{x}\|$, $\mathbf{x}^H$ and $(\mathbf{x})_n$ denote the transpose, Euclidean norm, conjugate transpose, n-th entry a vector x. diag$\{\mathbf{x}\}$ denotes a diagonal matrix with vector x as the main diagonal. "$\otimes$" denotes the Kronecker product.

II. System Model and Problem Formulation

A. System Model

As shown in Fig. 1, we consider the downlink transmission from a multi-antenna BS to a single-antenna legitimate user (Bob) aided by an AIRS in the presence of a single-antenna eavesdropper (Eve). We assume that the direct link between the BS and Bob/Eve is severely blocked such that the confidential signal can only be transmitted via the cascaded BS-AIRS-Bob/-Eve link. For convenience, we establish a global Cartesian coordinate system (CCS) in Fig. 1 and assume that the BS is located at the origin, with its coordinate denoted as $\mathbf{s} = [0, 0, 0]^T$. Moreover, we assume that Bob is located on the x-axis and denote its coordinate as $\mathbf{p}_B = [x_B, 0, 0]^T$. Let the coordinates of Eve and the UAV/AIRS be denoted as $\mathbf{p}_E = [x_E, y_E, 0]^T$ and $\mathbf{q} = [q_x, q_y, H]^T$, respectively, where H denotes the altitude of the UAV/AIRS that is assumed to be fixed, while both q_x and q_y can be flexibly adjusted by leveraging the flexible deployment of the UAV. The BS is assumed to be equipped with a uniform linear array (ULA) with M antennas, with the inter-antenna spacing denoted as d_B. The AIRS is assumed to be equipped with a uniform planar array with N reflecting elements, with the inter-element spacing denoted as d_I.

To characterize the rotation of the AIRS, we also establish a local CCS on it, with $\mathbf{q}$ located at its origin and the AIRS parallel to the x'-y' plane, as shown in Fig. 1. Let N_x and N_y denote the numbers of AIRS reflecting elements in these two dimensions, respectively. Define $\mathbf{\Phi} = \text{diag}\{e^{jw_1}, ..., e^{jw_N}\} \in \mathbb{C}^{N\times N}$ as the diagonal reflection matrix of the AIRS, where w_n denotes the phase shift of the n-th reflecting element. Let $\boldsymbol{\psi} = [\psi_z, \psi_y, \psi_x]^T$ denote the rotational angle vector of the AIRS, where ψ_i represents its rotational angle around the i-axis of the global CCS in radian, $i \in \{x, y, z\}$. Notably, the relationship between the global CCS and the local CCS can be characterized by the following matrices [13]

$$\mathbf{Q}(\boldsymbol{\psi}) = \mathbf{Q}_x(\psi_x)\mathbf{Q}_y(\psi_y)\mathbf{Q}_z(\psi_z), \tag{1}$$

where $\mathbf{Q}_x(\psi_x)$ indicates the rotation around x-axis and is given by

$$\mathbf{Q}_x(\psi_x) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\psi_x) & -\sin(\psi_x) \\ 0 & \sin(\psi_x) & \cos(\psi_x) \end{bmatrix}, \tag{2}$$

$\mathbf{Q}_y(\psi_y)$ indicates the orientation around the y-axis and is given by

$$\mathbf{Q}_y(\psi_y) = \begin{bmatrix} \cos(\psi_y) & 0 & \sin(\psi_y) \\ 0 & 1 & 0 \\ -\sin(\psi_y) & 0 & \cos(\psi_y) \end{bmatrix}, \tag{3}$$

and $\mathbf{Q}_z(\psi_z)$ indicates the orientation around the z-axis and is given by

$$\mathbf{Q}_z(\psi_z) = \begin{bmatrix} \cos(\psi_z) & -\sin(\psi_z) & 0 \\ \sin(\psi_z) & \cos(\psi_z) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{4}$$

Based on (1), for any given 3D location $\mathbf{p}$, its corresponding coordinate in the local CCS is given by

$$\mathbf{p}^{local} = \mathbf{Q}^T(\boldsymbol{\psi})(\mathbf{p} - \mathbf{q}). \tag{5}$$

As such, the coordinate of the BS in the local CCS is

$$\mathbf{s}^{local} = \mathbf{Q}^T(\boldsymbol{\psi})(\mathbf{s} - \mathbf{q}) = [s_x^{local}, s_y^{local}, s_z^{local}]^T, \tag{6}$$

and the coordinates of Bob and Eve in the local CCS are

$$\mathbf{p}_U^{local} = \mathbf{Q}^T(\boldsymbol{\psi})(\mathbf{p}_U - \mathbf{q}) \tag{7}$$

$$= [x_U^{local}, y_U^{local}, z_U^{local}]^T, U \in \{B, E\}. \tag{8}$$

Given the above local coordinates, the angle of departure (AoD) from the BS to the AIRS, the elevation/azimuth AoD from the AIRS to Bob/Eve, and the elevation/azimuth angle of arrival (AoA) at the AIRS from the BS can be respectively obtained as

$$\phi_{BA} = \arccos \frac{H}{\|\mathbf{q}\|}, \tag{9}$$

$$\phi_{AU}^{(e)} = \arccos \frac{z_U^{local}}{\|\mathbf{p}_U - \mathbf{q}\|}, \quad \phi_{AU}^{(a)} = \arctan \frac{y_U^{local}}{x_U^{local}}, \tag{10}$$

$$\theta_{AB}^{(e)} = \arccos \frac{s_z^{local}}{\|\mathbf{q}\|}, \quad \theta_{AB}^{(a)} = \arctan \frac{s_y^{local}}{s_x^{local}}, \tag{11}$$

where $U \in \{B, E\}$. Instead of adopting the conventional isotropic reflection model, we consider a more practical angle-dependent reflection model [7] in this paper. In particular, as shown in Fig. 1, denote by ϕ_1 and ϕ_{2U} the incident angle from the BS to the AIRS and the reflected angle from the AIRS to Bob/Eve, respectively, which can be calculated as

$$\phi_1 = \arccos \frac{-s_z^{local}}{\|\mathbf{q}\|}, \ \phi_{2U} = \arccos \frac{z_U^{local}}{\|\mathbf{p}_U - \mathbf{q}\|}, \tag{12}$$

where $U \in \{B, E\}$. Note that to ensure that BS and Bob are both located within the reflection half-space of the AIRS, it must hold that $\phi_1, \phi_{2B} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. In contrast, it may hold that $\phi_{2E} \in \left[-\pi, -\frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$, under which Eve cannot receive the reflected signal by the AIRS, thus resulting in *spacing nulling*. Based on the above, we can define the AIRS's effective aperture gain for Bob as [7]

$$F(\mathbf{p}_B, \boldsymbol{\psi}) = \cos\phi_1 \cos\phi_{2B}. \tag{13}$$

On the other hand, the effective aperture gain for Eve is a piecewise function depending on whether Eve is located in the reflection half-space, i.e.,

$$F(\mathbf{p}_E, \boldsymbol{\psi}) = \begin{cases} \cos\phi_1 \cos\phi_{2E}, & \phi_{2E} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \\ 0, & \phi_{2E} \in \left[-\pi, -\frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right). \end{cases} \tag{14}$$

By substituting (1)-(8) into (12) and after some manipulations, we can obtain

$$\cos\phi_1 = \frac{q_x L_1 + q_y L_2 + H L_3}{\sqrt{q_x^2 + q_y^2 + H^2}}, \tag{15}$$

$$\cos\phi_{2B} = \frac{(q_x - x_B)L_1 + q_y L_2 + H L_3}{\sqrt{(q_x - x_B)^2 + q_y^2 + H^2}} \tag{16}$$

and

$$\cos\phi_{2E} = \frac{(q_x - x_E)L_1 + (q_y - y_E)L_2 + HL_3}{\sqrt{(q_x - x_E)^2 + (q_y - y_E)^2 + H^2}}, \tag{17}$$

where $L_1 = \sin\psi_y$, $L_2 = -\cos\psi_y \sin\psi_x$ and $L_3 = \cos\psi_x \cos\psi_y$.

Let $\mathbf{L} = [L_1, L_2, L_3]^T$. Notably, we can always retrieve the AIRS rotational angles from any given $\mathbf{L}$. Then, we can represent (15)-(17) as

$$\cos\phi_1 = \frac{\mathbf{q}^T\mathbf{L}}{\|\mathbf{q}\|},$$

$$\cos\phi_{2B} = \frac{(\mathbf{q}-\mathbf{p}_B)^T\mathbf{L}}{\|\mathbf{q}-\mathbf{p}_B\|},\ \cos\phi_{2E} = \frac{(\mathbf{q}-\mathbf{p}_E)^T\mathbf{L}}{\|\mathbf{q}-\mathbf{p}_E\|}. \tag{18}$$

For $\phi_1, \phi_{2B} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ to hold, the following inequalities should be met:

$$\mathbf{q}^T\mathbf{L} \geq 0, \quad (\mathbf{q}-\mathbf{p}_B)^T\mathbf{L} \geq 0. \tag{19}$$

Moreover, in order to achieve space nulling at Eve, i.e., $\phi_{2E} \in \left[-\pi, -\frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$, it should hold that

$$(\mathbf{p}_E - \mathbf{q})^T\mathbf{L} \geq 0. \tag{20}$$

Hence, (13) and (14) can be respectively simplified as

$$F(\mathbf{p}_B, \boldsymbol{\psi}) = \frac{\mathbf{L}^T\mathbf{q}(\mathbf{q}-\mathbf{p}_B)^T\mathbf{L}}{\|\mathbf{q}\|\|\mathbf{q}-\mathbf{p}_B\|}, \tag{21}$$

$$F(\mathbf{p}_E, \boldsymbol{\psi}) = \begin{cases} \frac{\mathbf{L}^T\mathbf{q}(\mathbf{q}-\mathbf{p}_E)^T\mathbf{L}}{\|\mathbf{q}\|\|\mathbf{q}-\mathbf{p}_E\|}, & (\mathbf{q}-\mathbf{p}_E)^T\mathbf{L} > 0, \\ 0, & (\mathbf{p}_E - \mathbf{q})^T\mathbf{L} \geq 0. \end{cases} \tag{22}$$

In this paper, we focus on the performance optimization for the AIRS and the BS under the space nulling at Eve. Hence, the second case in (22) should hold. Due to the high altitude of the AIRS and the UAV to maintain the line-of-sight (LoS) channels with the ground nodes, we assume LoS BS-AIRS and AIRS-Bob channels[1]. Accordingly, the path loss over the BS-AIRS link can be obtained as $\beta_{BA} = \frac{\beta_0}{\|\mathbf{q}\|^2}$, where β_0 denotes the reference path loss at the reference distance of one meter (m), Furthermore, let $\beta_{AB} = \frac{\beta_0}{\|\mathbf{q}-\mathbf{p}_B\|^2}$ denote the path loss over the AIRS-Bob link. Next, to characterize the BS-AIRS and AIRS-Bob channels, we define the array response for a ULA as

$$\boldsymbol{\alpha}(n, \vartheta, d) = [1, e^{-j\frac{2\pi}{\lambda}\vartheta d}, ..., e^{-j\frac{2\pi}{\lambda}\vartheta(n-1)d}]^T, \tag{23}$$

where λ denotes the wavelength, ϑ denotes the directional cosine of the target direction, n denotes the antenna number, and d denotes the inter-antenna spacing. Moreover, the transmit array response at the BS and the receive array response at the AIRS can be respectively expressed as

$$\mathbf{a}_t(\phi_{BA}) = \boldsymbol{\alpha}(N_x, \cos(\phi_{BA}), d_B), \tag{24}$$

$$\mathbf{a}_r(\theta_{AB}^{(e)}, \theta_{AB}^{(a)}) = \boldsymbol{\alpha}(N_x, \sin\theta_{AB}^{(e)}\cos\theta_{AB}^{(a)}, d_I) \otimes \boldsymbol{\alpha}(N_y, \sin\theta_{AB}^{(e)}\sin\theta_{AB}^{(a)}, d_I). \tag{25}$$

The array response in the AIRS-Bob link is represented as

$$\mathbf{a}_B(\phi_{AB}^{(e)}, \phi_{AB}^{(a)}) = \boldsymbol{\alpha}(N_x, \sin\phi_{AB}^{(e)}\cos\phi_{AB}^{(a)}, d_I) \otimes \boldsymbol{\alpha}(N_y, \sin\phi_{AB}^{(e)}\sin\phi_{AB}^{(a)}, d_I), \tag{26}$$

[1]Note that the AIRS-Eve channel can be of any type due to the space nulling at Eve.

Based on the above array responses, the channel from the BS to the AIRS is given by

$$\mathbf{H}_{BA} = \sqrt{\beta_{BA}}e^{-j\frac{2\pi\|\mathbf{q}\|}{\lambda}}\mathbf{a}_r(\theta_{AB}^{(e)}, \theta_{AB}^{(a)})\mathbf{a}_t^H(\phi_{BA}), \tag{27}$$

and that from the AIRS to Bob is

$$\mathbf{h}_{AB}^H = \sqrt{\beta_{AB}}e^{-j\frac{2\pi\|\mathbf{q}-\mathbf{p}_B\|}{\lambda}}\mathbf{a}_B^H(\phi_{AB}^{(e)}, \phi_{AB}^{(a)}). \tag{28}$$

Therefore, the received signal at Bob can be expressed as

$$y_B = \mathbf{h}_{AB}^H\mathbf{\Phi}\mathbf{H}_{BA}\boldsymbol{v}\sqrt{F(\mathbf{p}_B, \boldsymbol{\psi})P}s + n_B. \tag{29}$$

where $\boldsymbol{v} \in \mathbb{C}^{N\times 1}$, P and s denote the BS's beamforming vector, transmit power and data symbol, respectively, and $n_B \sim \mathcal{N}(0, \sigma^2)$ is the additive white Gaussian noise (AWGN) with zero mean and variance of σ^2. Thus, the received signal-to-noise ratio (SNR) at Bob can be expressed as

$$\begin{aligned}\gamma_B =& \frac{P\beta_0^2F(\mathbf{p}_B, \boldsymbol{\psi})|\mathbf{h}_{AB}^H\mathbf{\Phi}\mathbf{H}_{BA}\boldsymbol{v}|^2}{\|\mathbf{q}\|^2\|\mathbf{q}-\mathbf{p}_B\|^2\sigma^2} \\ =& \frac{\bar{P}\beta_0^2F(\mathbf{p}_B, \boldsymbol{\psi})|\mathbf{f}_B^H\boldsymbol{\omega}|^2|\mathbf{a}_t^H(\phi_{BA})\boldsymbol{v}|^2}{\|\mathbf{q}\|^2\|\mathbf{q}-\mathbf{p}_B\|^2},\end{aligned} \tag{30}$$

where $\bar{P} = \frac{P}{\sigma^2}$,

$$\mathbf{f}_B = \boldsymbol{\alpha}(N_x, \varphi_x, d_I) \otimes \boldsymbol{\alpha}(N_y, \varphi_y, d_I), \tag{31}$$

with $\varphi_x = \sin\phi_{AB}^{(e)}\cos\phi_{AB}^{(a)} - \sin\theta_{AB}^{(e)}\cos\theta_{AB}^{(a)}$, $\varphi_y = \sin\phi_{AB}^{(e)}\sin\phi_{AB}^{(a)} - \sin\theta_{AB}^{(e)}\sin\theta_{AB}^{(a)}$, and $\boldsymbol{\omega} = [e^{jw_1}, ..., e^{jw_N}]^T$ denotes the passive beamforming vector of the AIRS.

Based on (30) and the space nulling, the secrecy rate of the considered system is equal to the achievable rate at Bob, i.e.,

$$\begin{aligned} R_s =& \log_2(1+\gamma_B) \\ =& \log_2\left(1 + \frac{\bar{P}\beta_0^2F(p_B, \boldsymbol{\psi})}{||\mathbf{q}||^2||\mathbf{q}-\mathbf{p}_B||^2}|\mathbf{f}_B^H\boldsymbol{\omega}|^2|\mathbf{a}_B^H(\phi_{BA})\boldsymbol{v}|^2\right).\end{aligned} \tag{32}$$

It can be easily seen that to maximize R_s, the maximum ratio transmission (MRT) should be adopted at the BS, i.e., $\boldsymbol{v}^* = \frac{\mathbf{a}_B(\phi_{BA})}{\sqrt{M}}$. Furthermore, the optimal passive beamforming should be set as $\boldsymbol{\omega}^* = \mathbf{f}_B$ to maximize $|\mathbf{f}_B^H\boldsymbol{\omega}|^2$. By substituting $\boldsymbol{v}^*$ and $\boldsymbol{\omega}^*$ into (32), the secrecy rate can be simplified as

$$\begin{aligned} R_s =& \log_2\left(1 + \bar{P}\beta_0^2MN^2\frac{F(\mathbf{p}_B, \boldsymbol{\psi})}{||\mathbf{q}||^2||\mathbf{q}-\mathbf{p}_B||^2}\right) \\ =& \log_2\left(1 + \bar{P}\beta_0^2MN^2\frac{\mathbf{L}^T\mathbf{S}_B\mathbf{L}}{||\mathbf{q}||^3||\mathbf{q}-\mathbf{p}_B||^3}\right),\end{aligned} \tag{33}$$

where we have defined $\mathbf{S}_B = \mathbf{q}(\mathbf{q}-\mathbf{p}_B)^T$.

B. Problem Formulation

Our goal in this paper is to maximize the secrecy rate in (33) by jointly optimizing the AIRS's location $\mathbf{q}$ and its 3D rotation angles $\boldsymbol{\psi}$ subject to the space nulling constraint at Eve. Therefore, the problem can be formulated as

$$\text{(P1)} \max_{\mathbf{q},\boldsymbol{\psi}} \quad \frac{\mathbf{L}^T\mathbf{S}_B\mathbf{L}}{\|\mathbf{q}\|^3\|\mathbf{q}-\mathbf{p}_B\|^3} \tag{34}$$

$$\text{s.t.} \quad \mathbf{q} \in \mathcal{Q}, \tag{35}$$

$$\mathbf{q}^T\mathbf{L} \geq 0, \tag{36}$$

$$(\mathbf{q}-\mathbf{p}_B)^T\mathbf{L} \geq 0, \tag{37}$$

$$(\mathbf{p}_E - \mathbf{q})^T\mathbf{L} \geq 0, \tag{38}$$

where $\mathcal{Q}$ denotes a prescribed region for the AIRS's movement, and we have dropped the logarithmic operator and constant term $\bar{P}\beta_0^2MN^2$ in the objective function of (P1).

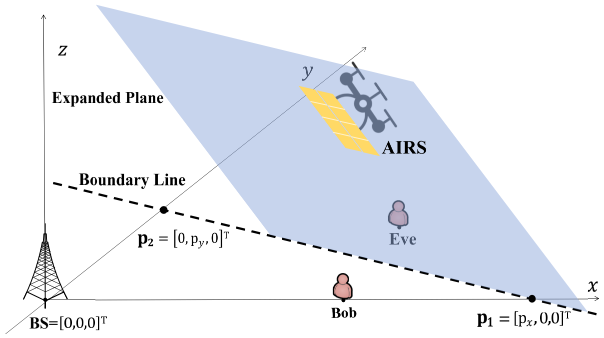

Fig. 2. Illustration of the intersection line between the AIRS and the ground.

Regarding (P1), we have the following proposition to characterize its feasibility.

Proposition 1: The space nulling is always feasible unless Eve is located on the line segment between the BS and Bob.

Proof: To prove this proposition, we expand the AIRS and it intersects the ground with a straight line. If Eve is not located on the BS-Bob line segment, we can always find a straight line between Eve and the BS-Bob segment, as illustrated in Fig. 2. Then, by placing the AIRS within any plane going through this straight line and orienting it to the side of the BS and Bob, we can always achieve signal nulling at Eve. However, if Eve is located on the BS-Bob segment, Eve must be located at the same side as the BS and/or Bob, rendering space nulling infeasible. This completes the proof. ■

However, (P1) is a non-convex optimization problem with the optimization variables $\mathbf{q}$ and $\boldsymbol{\psi}$ intricately coupled in the objective function and constraints. In the next section, we propose an efficient method to solve it.

III. Proposed Solution To (P1)

To solve (P1), we first show that for any given AIRS/UAV position, it admits an optimal AIRS rotation in closed-form.

A. Rotation Optimization for any Given Location

For any given location $\mathbf{q}$ of the UAV, (P1) reduces to

$$\text{(P1.1)} \ \max_{\boldsymbol{\psi}} \quad \mathbf{L}^T\mathbf{S}_B\mathbf{L} \tag{39}$$

$$\text{s.t.} \quad \mathbf{q}^T\mathbf{L} \geq 0, \tag{40}$$

$$(\mathbf{q}-\mathbf{p}_B)^T\mathbf{L} \geq 0, \tag{41}$$

$$(\mathbf{p}_E-\mathbf{q})^T\mathbf{L} \geq 0, \tag{42}$$

where

$$\begin{aligned}\mathbf{L}^T\mathbf{S}_B\mathbf{L} &= \begin{bmatrix} L_1 \\ L_2 \\ L_3 \end{bmatrix}^T \begin{bmatrix} q_x^2 - q_x x_B & q_x q_y & q_x H \\ q_x q_y - q_y x_B & q_y^2 & q_y H \\ q_x H - H x_B & q_y H & H^2 \end{bmatrix} \begin{bmatrix} L_1 \\ L_2 \\ L_3 \end{bmatrix} \\ &= L_1^2 \Bigg(q_x^2 + 2q_x q_y \left(\frac{L_2}{L_1}\right) + 2q_x H \frac{L_3}{L_1} + q_y^2 \left(\frac{L_2}{L_1}\right)^2 \\ &\quad + 2q_y H \left(\frac{L_2}{L_1}\right)\left(\frac{L_3}{L_1}\right) + H^2 \left(\frac{L_3}{L_1}\right)^2 \Bigg) \\ &\quad - L_1^2 x_B \left(q_x + q_y \left(\frac{L_2}{L_1}\right) + H \left(\frac{L_3}{L_1}\right) \right). \end{aligned} \tag{43}$$

Next, we show (P1.1) can be equivalently recast as a simplifed version with respect to (w.r.t.) two scalars only. To this end, as shown in Fig. 2, we expand the AIRS to form a plane defined by

$$(\mathbf{q}-\mathbf{p})^T\mathbf{L} = 0, \ \{\mathbf{p}_1, \mathbf{p}_2\} \in \mathbf{p} \tag{44}$$

and it intersects with the ground at a straight line $\mathbf{r}$ i.e.,

$$\mathbf{r} = \lambda\mathbf{p}_1 + (1-\lambda)\mathbf{p}_2, \ \lambda \in \mathbb{R}, \tag{45}$$

where $\mathbf{p}_1 = [p_x, 0, 0]^T$ and $\mathbf{p}_2 = [0, p_y, 0]^T$, with p_x and p_y denoting the intersection of the expanded plane with x- and y-axes, which are respectively given by

$$p_x = q_x + q_y \frac{L_2}{L_1} + H\frac{L_3}{L_1}, \tag{46}$$

$$p_y = \frac{L_1}{L_2} q_x + q_y + H\frac{L_3}{L_2} = \frac{L_1}{L_2} p_x. \tag{47}$$

Based on Proposition 1, the straight line in (45) results in a boundary between Bob and Eve, which should be located at its different sides. It should also be mentioned that each AIRS rotation solution yields a different boundary line. As any straight line can be uniquely characterized by its intercept and slope, it is equivalent to optimize the x-intercept and slope of (45) to determine the AIRS rotation angles, namely, p_x and $T \triangleq p_x/p_y$. By noting

$$L_2/L_1 = T, \quad L_3/L_1 = \frac{(p_x - q_x) - q_y T}{H}, \tag{48}$$

and $L_1^2 + L_2^2 + L_3^2 = 1$, the objective function of (P1.1) can be recast as a form in terms of p_x and T only, i.e.,

$$\mathbf{L}^T\mathbf{S}_B\mathbf{L} = L_1^2 p_x (p_x - x_B), \tag{49}$$

with

$$L_1^2 = \frac{H^2}{H^2 + (HT)^2 + (p_x - q_x - q_y T)^2}. \tag{50}$$

Furthermore, for constraint (40) in (P3), we have $p_x L_1 = \mathbf{q}^T\mathbf{L} \geq 0$. While for (41), we have $(\mathbf{q}-\mathbf{p}_B)^T\mathbf{L} = p_x L_1 - x_B L_1 \geq 0$. Given that p_x and L_1 have the same sign, (41) is equivalent to $p_x(p_x - x_B) \geq 0$. Similarly, it can be shown that (42) is equivalent to $p_x(x_E - p_x + y_E T) \geq 0$.

Based on the above, (P1.1) can be equivalently recast as the following lower-dimensional optimization problem,

$$\text{(P1.2)} \min_{p_x, T} \quad \frac{H^2 + (HT)^2 + (p_x - q_x - q_y T)^2}{p_x (p_x - x_B)} \tag{51}$$

$$\text{s.t.} \quad p_x L_1 \geq 0, \tag{52}$$

$$p_x(p_x - x_B) \geq 0, \tag{53}$$

$$p_x(x_E - p_x + y_E T) \geq 0. \tag{54}$$

For (P1.2), we can derive its optimal solution in closed-form. Note that for any given p_x, the nominator of (51) is a quadratic function of T, while constraint (54) sets a lower bound on it, which is given by $T_{\text{lb}} = \frac{p_x - x_E}{y_E}$. In the absence of this lower bound, the optimal solution for T can be obtained as $T_0 = \frac{(p_x - q_x) q_y}{H^2 + q_y^2}$ by taking the derivative of the numerator of (51) w.r.t. T. Then, by leveraging the property of the quadratic function, it is not difficult to see the optimal T for (P1.2) is given by

$$T^*(p_x) = \begin{cases} \frac{(p_x - q_x) q_y}{H^2 + q_y^2}, & \text{if} \quad p_x(x_E - p_x + y_E T_0) \geq 0, \\ \frac{p_x - x_E}{y_E}, & \text{otherwise.} \end{cases} \tag{55}$$

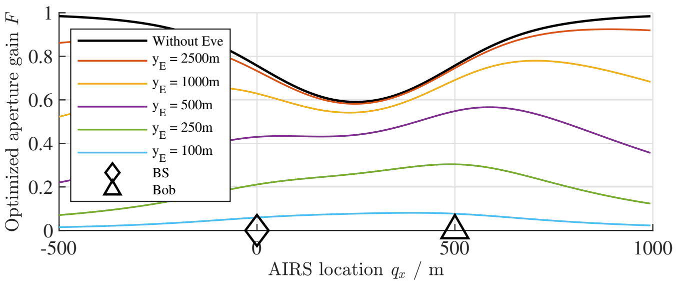

Fig. 3. Optimized aperture gain $F(\mathbf{p}_B, \boldsymbol{\psi})$ versus the AIRS's location with $\mathbf{q} = [q_x, 0, 300]^T$ and $\mathbf{q}_E = [400, y_E, 0]^T$.

Substituting (55) into (P1.2) yields a further simplified optimization problem w.r.t. p_x only, i.e.,

$$\text{(P1.3)} \min_{p_x} \frac{H^2 + (HT^*(p_x))^2 + (p_x - q_x - q_y T^*(p_x))^2}{p_x(p_x - x_B)}$$
$$\text{s.t. } p_x L_1 \geq 0, \quad (56)$$
$$p_x(p_x - x_B) \geq 0, \quad (57)$$
$$p_x(x_E - p_x + y_E T_0) \geq 0, \quad (58)$$

and its optimal solution is given by

$$p_x^* = \begin{cases} \frac{\|\mathbf{q}\|^2}{2q_x - x_B} + \frac{\sqrt{\|\mathbf{q}\|^4 - x_B(2q_x - x_B)\|\mathbf{q}\|^2}}{|2q_x - x_B|}, \\ \qquad \text{if } p_x^*(x_E - p_x^* + y_E T_0) \geq 0, \\ \frac{C}{-x_B A - B} + \frac{\sqrt{C^2 + x_B(x_B A + B)C}}{|-x_B A - B|}, \\ \qquad \text{otherwise}, \end{cases} \quad (59)$$

where $A = H^2 + (q_y - y_E)^2$, $B = -2(q_y - y_E)(q_y x_E - q_x y_E) - 2H^2 x_E$ and $C = H^2(x_E^2 + y_E^2) + (q_y x_E - q_x y_E)^2$. The details of deriving (59) are provided in Appendix A.

It should be mentioned that the first case of (59) corresponds to the optimal AIRS rotation in the absence of Eve (i.e., without constraint (58)). Hence, the corresponding p_x^* is independent of Eve's location. However, if this rotation solution fails to exclude Eve from the reflection half-space, the optimal p_x needs to account for Eve's location to achieve space nulling, resulting in the second case in (59).

With the optimized p_x^* and T^*, we can retrieve the optimal $\mathbf{L}$. Specifically, based on (50), we can obtain the optimal L_1 as $L_1^* = \pm 1/\sqrt{(1 + (T^*)^2 + (p_x^* - q_x + \frac{q_y}{H} T^*)^2)}$. Subject to the constraint (52), we can determine the sign of L_1^* by $\frac{p_x^*}{|p_x^*|}$. By noting (48), we have $L_2^* = L_1^* T^*$, and L_3^* can be obtained from $L_1^2 + L_2^2 + L_3^2 = 1$. Finally, we can retrive the optimal rotational angles from L_i^*, $i = 1, 2, 3$ as

$$\begin{cases} \psi_x^* &= -\arcsin \frac{L_2^*}{\sqrt{1 - (L_1^*)^2}}, \\ \psi_y^* &= \arcsin L_1^*, \\ \psi_z^* &\in \mathbf{R}. \end{cases} \quad (60)$$

To gain more insights, based on the optimal rotation angles in (60), we plot the resultant Bob's aperture gain $F(\mathbf{p}_B, \boldsymbol{\psi}^*)$ in (13) versus the location of the AIRS on x-axis, i.e., q_x in Fig. 3. It is observed from Fig. 3 that the presence of Eve or space nulling inevitably results in a loss in the maximum aperture gain achievable for Bob. Nonetheless, this loss is observed to reduce as the distance from Eve to x-axis, i.e., y_E, increases. This means that space nulling can be more easily achieved if Bob and Eve are sufficiently separated without sacrificing Bob's maximum achievable aperture gain.

Following this, we further consider a special case that the AIRS is located above the x-axis, i.e., $q_y = 0$. In this case, we have $T = 0$ by noting the first case in (55). Then, the corresponding boundary line should be perpendicular to the x-axis based on (55). Furthermore, assuming that Eve is sufficiently far away from Bob, i.e., $y_E \to \infty$, we can obtain

$$\psi_y^*(q_x) = \frac{\pi - \psi_1(q_x) - \psi_2(q_x)}{2}, \quad (61)$$

where

$$\psi_1(q_x) = \arccos \frac{q_x}{\|\mathbf{q}\|}, \quad \psi_2(q_x) = \arccos \frac{q_x - x_B}{\|\mathbf{q} - \mathbf{p}_B\|}. \quad (62)$$

It is interesting to note that the optimal rotation in (61) is consistent with that derived in [10] without Eve. This, again, shows that the space nulling can achieve near-optimal performance in the case that Eve is remote from Bob.

B. Location Optimization

With the optimal rotational angles $\boldsymbol{\psi}$ obtained in Section III-A, problem (P1) can be simplified as

$$\text{(P2)} \max_{\mathbf{q}} \frac{\mathbf{L}(\mathbf{q})^T \mathbf{S}_B \mathbf{L}(\mathbf{q})}{\|\mathbf{q}\|^3 \|\mathbf{q} - \mathbf{p}_B\|^3} \quad (63)$$
$$\text{s.t.} \quad \mathbf{q} \in \mathcal{Q}. \quad (64)$$

However, (63) is highly non-convex in terms of $\mathbf{q}$, making it challenging to derive a closed-form optimal solution. To deal with this, we employ a two-dimensional search over q_x and q_y to determine the optimal $\mathbf{q}$.

To gain more insights into the optimal $\mathbf{q}$, we also consider the following special cases. First, if $H \to 0$, i.e., the AIRS becomes a terrestrial IRS or passive 6DMA [12], it can be shown that the first case in (59) is more easily achieved. As such, the optimal solution to (P2) should be closer to the optimal AIRS location in the absence of Eve. On the contrary, if $H \to +\infty$, we have $T_0 \to 0$ and $p_x^* \to \infty$. Hence, the second case of (59) is easier to be achieved, rendering the optimal solution farther away from the optimal AIRS position without Eve.

IV. Numerical Results

In this section, numerical results are provided to evaluate the performance of our proposed space-nulling approach. The coordinates of Bob and Eve are $\mathbf{p}_B = [500, 0, 0]^T$ and $\mathbf{p}_E = [400, y_E, 0]^T$, respectively, while the movement region of the AIRS is set as $\mathcal{Q} \in \{(q_x, q_y) | -100 < q_x < 700, -50 < q_y < 300\}$. Unless otherwise stated, the altitude of the AIRS is fixed as $H = 100$ m and the distance from Eve to x-axis is $y_E = 250$ m. The BS's transmit power and the noise power are set as P = 20 dBm and $\sigma^2 = -110$ dBm, respectively. The reference LoS path gain is $\beta_0 = -40$ dB. Furthermore, we set the number of BS antennas as M = 64, and the number of the AIRS reflecting elements as N = 900. We consider the following benchmark schemes for performance comparison:

- Achievable rate at Bob without Eve, serving as an upper bound of R_s,

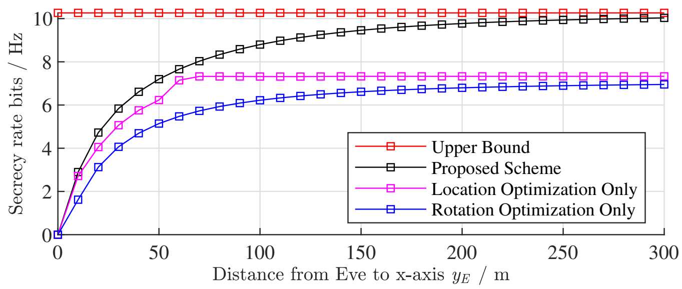

Fig. 4. Secrecy rate versus Eve's distance to x-axis y_E

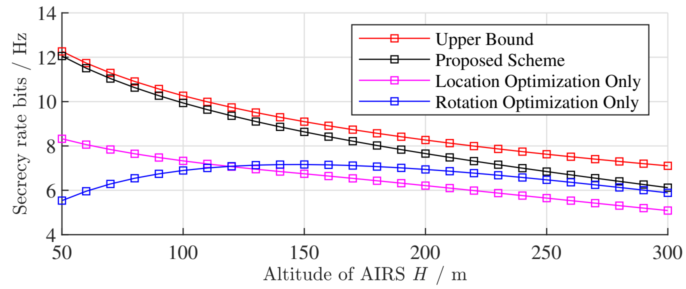

Fig. 5. Secrecy rate versus the AIRS's altitude H

• AIRS's rotation optimization only, with $\mathbf{q} = [250, 0, H]^T$ under the space-nulling constraint,

• AIRS's location optimization only, with $\boldsymbol{\psi} = [0, 0, -\frac{\pi}{2}]^T$ under the space-nulling constraint.

First, Fig. 4 shows the secrecy rates by the proposed scheme and the three benchmark schemes versus the distance from Eve to x-axis, i.e., y_E, under $H = 100$ m. It is observed from Fig. 4 that as y_E increases, the secrecy rate by the proposed scheme ultimately increases and gradually converges to the maximum achievable rate at Bob without Eve. This is consistent with our analytical results derived in Section III-A. However, in the extreme case of $y_E = 0$, Eve is located on the line segment between the BS and Bob, rendering space nulling infeasible and resulting in a zero secrecy rate. In this case, the AIRS passive beamforming may also be optimized taking into account both Bob and Eve, as will be pursued in our follow-up work. Moreover, the scheme with location optimization only is observed to outperform that with rotation optimization only, implying that the former may play a more significant role than the latter to achieve optimal space nulling.

Fig. 5 shows the secrecy rates by different schemes versus the AIRS's altitude H with $y_E = 250$ m. It is observed that as H increases, the performance of the proposed scheme and the upper bound degrades due to the more severe end-to-end path loss from the BS to Bob. It is also observed that the performance gap between the upper bound and the proposed scheme increases with H, while the gap between the proposed scheme and "rotation optimization only" decreases. These observations imply that space nulling is more effective at low-to-moderate altitudes than at higher altitudes. This is consistent with our analytical results presented in Section III-B. It is also observed that under the considered simulation setup, the rotation optimization may become dominant over position optimization at a high altitude to achieve optimal space nulling.

V. Conclusions

In this paper, we proposed a new space-nulling approach for secure communications by leveraging the UAV-mounted passive 6DMAs under the angle-dependent model. We derived the closed-form solution for the AIRS's optimal 3D rotational angles for any given location under the space-nulling constraint at Eve. Both analytical and numerical results showed that the proposed space-nulling scheme can achieve near-optimal performance if the altitude of the AIRS is not high and/or Eve is far away from the straight line formed by the BS and Bob.

Appendix A
Proof of the Proposed Solution to (59)

By substituting T^* in (55) into the objective function of (P1.2), it becomes $\frac{(p_x-q_x)^2+H^2+q_y^2}{p_x(p_x-x_B)}$ and $\frac{Ap_x^2+Bp_x+C}{p_x(p_x-x_B)}$ for the first and second cases, respectively, where $A = H^2 + (q_y - y_E)^2$, $B = -2(q_y - y_E)(q_y x_E - q_x y_E)$, $C = H^2(x_E^2 + y_E^2) + (q_y x_E - q_x y_E)^2$. For both cases, we can take the derivative w.r.t. p_x and obtain their corresponding optimal solutions as the first and second cases of (59), respectively.

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